COURSE OUTLINE

Data Science & AI

MODULE 01 · Duration: 5 Weeks

Python Foundations

01. Introduction, Installation, IDE Setup
Setting up Python, choosing an IDE, and running your first program.

02. Variables, Data Types, Operators
Understanding different data types, declaring variables, and using arithmetic, logical, and comparison operators.

03. Conditional Statements
Using if, elif, and else for decision-making in programs.

04. Loops & Iterations
Implementing repetitive tasks with for and while loops, including break and continue.

05. Functions, Lambda, Modules
Creating reusable code blocks, anonymous functions (lambda), and importing Python modules.

06. Python Data Structures
Working with lists, tuples, dictionaries, and sets to organize and manage data.

07. File Handling
Reading from and writing to files for persistent data storage.

08. Exception Handling
Managing errors gracefully using try, except, and finally.

09. Object-Oriented Programming
Introduction to classes, objects, inheritance, and encapsulation in Python.

CLASS-BASED PROJECTS

- Student Grading System – Implementing OOP concepts to calculate and manage student grades.
- Salary/Billing System (OOP) – Designing a billing system using classes and objects.

HOME-BASED PROJECTS

- Employee Record System – File handling project to store and manage employee details.
- Library Management System (File Handling) – Creating a system to manage books and members using file operations.

MODULE 02 · Duration: TBD

Python for Data Science (NumPy, Pandas, Visualization)

01. NumPy Core Operations
Learn numerical computing with arrays, vectorized operations, and mathematical functions for efficient data handling.

02. Pandas DataFrames
Introduction to tabular data structures, creating, indexing, and manipulating DataFrames.

03. Data Cleaning & Transformation
Techniques to clean, normalize, and prepare data for analysis.

04. Handling Missing Values
Identifying and managing missing or null data in datasets.

05. Merging, GroupBy
Combining datasets and performing aggregation operations for analysis.

06. Data Visualization (Matplotlib, Seaborn, Plotly)
Creating charts, plots, and interactive visualizations to explore and present data effectively.

CLASS-BASED PROJECTS

- COVID Dataset Exploratory Data Analysis (EDA) – Analyze COVID-19 data to extract insights and trends.
- Sales Dashboard using Pandas + Plotly – Build an interactive dashboard for visualizing sales performance.

HOME-BASED PROJECTS

- E-commerce Customer EDA – Analyze customer behavior and purchase patterns using real-world datasets.
- Weather Data Analysis Notebook – Perform analysis and visualization of historical weather data.

MODULE 03 · Duration: 2 Weeks

Statistics & Math for ML

01. Descriptive Statistics
Summarizing and describing data using measures like mean, median, mode, variance, and standard deviation.

02. Probability & Distributions
Understanding probability theory and common distributions (Normal, Binomial, Poisson) used in ML.

03. Hypothesis Testing
Performing statistical tests to make data-driven decisions and validate assumptions.

04. Covariance & Correlation
Measuring relationships and dependencies between variables.

05. Linear Algebra Basics
Key concepts like vectors, matrices, and operations essential for machine learning algorithms.

06. Cost Functions & Gradients

Understanding loss functions and gradient-based optimization for model training.

CLASS-BASED PROJECTS

- Company Sales Statistical Report – Analyze company sales data using descriptive statistics.
- Correlation Study (Any Dataset) – Study variable relationships and correlations within a dataset.

HOME-BASED PROJECTS

- Hypothesis Testing Assignment – Apply statistical tests to real datasets to validate hypotheses.
- Finance/Stock Statistical Exploration – Explore financial data using descriptive statistics and correlation analysis.

MODULE 04 · Duration: 8 Weeks

Machine Learning

01. ML Pipeline

Understanding the end-to-end machine learning workflow from data collection to model deployment.

02. Feature Engineering

Techniques to create, transform, and select meaningful features to improve model performance.

03. Data Preprocessing

Handling missing values, encoding categorical variables, and scaling features for modeling.

04. Train/Test Split & Cross Validation

Dividing datasets to evaluate model performance and avoid overfitting.

05. Supervised Learning

Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting (XGBoost/LightGBM), KNN, and Support Vector Machine.

06. Unsupervised Learning

K-Means Clustering, Hierarchical Clustering, and Principal Component Analysis (PCA).

07. Model Evaluation and Optimization

Classification and regression metrics, and hyperparameter tuning.

CLASS-BASED PROJECTS

- House Price Prediction – Predict housing prices using regression algorithms.
- Customer Segmentation (K-Means) – Group customers into segments based on behavior and characteristics.
- Loan Eligibility Prediction – Classify loan applicants as eligible or not using supervised learning.
- Fraud Detection Model – Detect fraudulent transactions using classification techniques.

HOME-BASED PROJECTS

- Customer Churn Prediction – Predict customer attrition using real-world datasets.
- Sales Forecasting – Forecast sales trends using machine learning models.
- Credit Risk Scoring Model – Assess the creditworthiness of clients using predictive modeling.
- HR Attrition Prediction – Predict employee turnover using historical HR data.

MODULE 05 · Duration: 6 Weeks

Deep Learning

01. Neural Network Fundamentals

Perceptron, Forward & Backpropagation – understanding the basic unit of neural networks and how information flows and updates weights during learning.

02. Frameworks (TensorFlow/Keras or PyTorch)

Training, validation, and callbacks for building and optimizing neural networks.

03. Computer Vision

Convolutional Neural Networks (CNNs) for image data, and Transfer Learning using pre-trained models like ResNet and VGG.

04. Natural Language Processing (NLP)

Text Cleaning, Word Embeddings, and sequential models: Recurrent Neural Networks (RNN), LSTM, and GRU.

CLASS-BASED PROJECTS

- CNN-Based Image Classifier – Classify images using convolutional neural networks.
- LSTM Movie Review Sentiment Analyzer – Predict sentiment from movie reviews using LSTM networks.
- Transfer Learning (ResNet/VGG) – Apply pre-trained models to new image classification tasks.

HOME-BASED PROJECTS

- Dogs vs Cats Classification – Build a model to classify dog and cat images.
- Spam Detection using Deep Learning – Detect spam messages or emails using neural networks.

MODULE 06 · Duration: 4 Weeks

MLOps

01. ML Deployment Concepts

Understanding how to deploy machine learning models into production environments.

02. FastAPI for ML Deployment

Building APIs to serve ML models for real-time predictions.

03. Docker Containers

Containerizing ML applications for consistent and scalable deployment.

04. MLflow for Tracking

Tracking experiments, metrics, and models to manage the ML lifecycle.

05. CI/CD Introduction

Introduction to Continuous Integration and Continuous Deployment for ML pipelines.

06. Logging & Monitoring

Implementing logs and monitoring to track model performance in production.

CLASS-BASED PROJECTS

- Build and Deploy an ML Model with FastAPI – Create an API to serve predictions from a trained model.
- MLflow Tracking Implementation – Track experiments, model metrics, and versions using MLflow.

HOME-BASED PROJECTS

- Dockerized ML Microservice – Package a machine learning model as a Docker container for deployment.
- Full Deployment Pipeline with Monitoring – Build an end-to-end ML pipeline with deployment and monitoring capabilities.

MODULE 07 · Duration: 5 Weeks

Big Data & Cloud

01. Big Data (Hadoop)

Hadoop framework concepts including HDFS for distributed storage, MapReduce for processing, and YARN for resource management.

02. Apache Spark

RDDs, DataFrames, Spark SQL, and Spark MLlib for scalable data processing and machine learning.

03. Cloud (AWS or Azure)

S3 Storage, EC2 Basics, Databricks/EMR, and deploying ML models on cloud infrastructure.

CLASS-BASED PROJECTS

- ETL Pipeline using PySpark – Build end-to-end data pipelines for extraction, transformation, and loading.
- Spark MLlib Model Training – Train and evaluate machine learning models using Spark MLlib.
- Cloud-Based Data Processing Workflow – Implement scalable data workflows on cloud platforms.

HOME-BASED PROJECTS

- Big Data Batch Processing – Perform batch data processing using Big Data tools.
- Cloud-Based Data Lake Setup – Set up and manage a cloud-based data lake for large datasets.
- Real-Time Processing Mini Project (Optional) – Explore real-time data processing techniques.

MODULE 08 · Duration: 4 Weeks

Generative AI (LLMs)

01. Transformers

Understand the core architecture behind modern language models.

02. BERT, GPT Conceptual Understanding

Learn the principles of popular NLP models like BERT and GPT.

03. Tokenization

Break text into tokens for model processing and understanding.

04. Prompt Engineering

Craft effective prompts to guide LLM responses.

05. Fine Tuning (LoRA/QLoRA)

Adapt pre-trained models to specific tasks using lightweight fine-tuning techniques.

06. RAG Systems
Implement Retrieval-Augmented Generation for context-aware AI responses.

07. Vector Databases (FAISS/Pinecone)
Store and search embeddings efficiently for semantic retrieval.

CLASS-BASED PROJECTS

- ChatGPT-style Q/A Bot – Build an interactive question-answering chatbot.
- RAG System with Vector Database – Develop a retrieval-augmented system using vector search.

HOME-BASED PROJECTS

- Domain-Fine-Tuned Q/A LLM – Fine-tune an LLM for specialized domain knowledge.
- Text Summarization or Content Generator – Create AI tools for summarizing or generating text content.

MODULE 09 · Duration: 4 Weeks

Agentic AI

01. What are AI Agents?
Introduction to autonomous AI agents and their capabilities.

02. Tool-Using Agents
Learn how agents interact with external tools to perform tasks.

03. Reasoning & Planning
Implement logic, decision-making, and multi-step task planning.

04. LangChain Agents
Build intelligent agents using the LangChain framework.

05. CrewAI & AutoGen
Explore multi-agent orchestration platforms for complex workflows.

06. Memory-Enabled Agents
Enable agents to store and recall context for smarter interactions.

07. Multi-Agent Systems
Coordinate multiple AI agents to solve collaborative tasks.

CLASS-BASED PROJECTS

- AI Autonomous Data Analyst – Build an AI agent to analyze and report on data automatically.
- Multi-Agent System (Researcher + Writer + Reviewer) – Develop a collaborative AI system with specialized agent roles.

HOME-BASED PROJECTS

- Personal AI Assistant with Tools – Create a custom AI assistant capable of executing tasks.
- Automated Email + Reporting Agent System – Automate communication and reporting using AI agents.